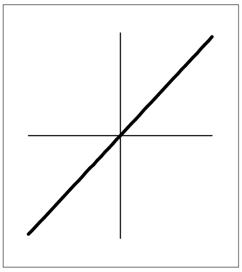
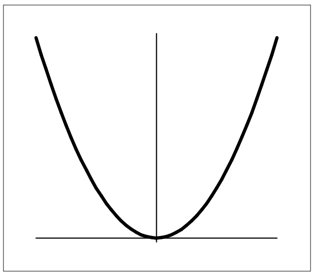
wtw158 Some important functions First sem 2011

Definition (p 29)
The function $y = f(x) = x^a$, with a a constant is called a **power function**

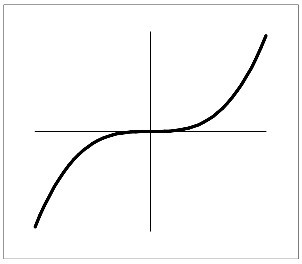
1. If $n = 1$ then $y = x$ is a linear function
2. If $n = 2, 4, 6, \ldots$ the shape of the graph $y = x^n$ is similar to the graph of $y = x^2$.
3. If $n = 3, 5, 7, \ldots$ the shape of the graph $y = x^n$ is similar to the graph of $y = x^3$.



$y = x$



$y = x^n, n = 2, 4, 6, \ldots$



$y = x^n, n = 3, 5, 7, \ldots$

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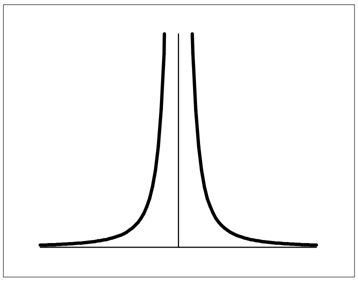
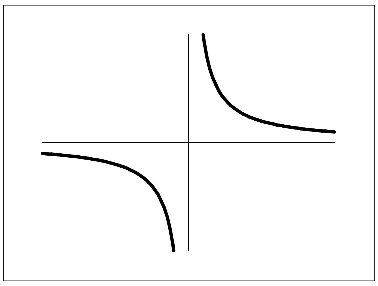
4. If $n = -2, -4, \ldots$ the shape of the graph $y = x^n$ is similar to the graph of $y = \frac{1}{x^2}$.
5. If $n = -1, -3, -5, \ldots$ the shape of the graph $y = x^n$ is similar to the graph of $y = \frac{1}{x}$.



$y = x^n, n = -2, -4, -6, \ldots$



$y = x^n, n = -1, -3, -5, -7, \ldots$

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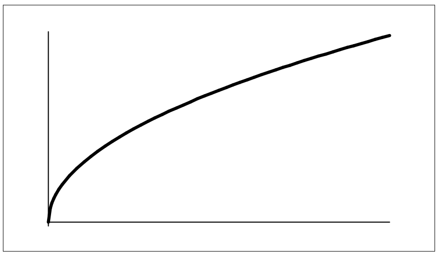
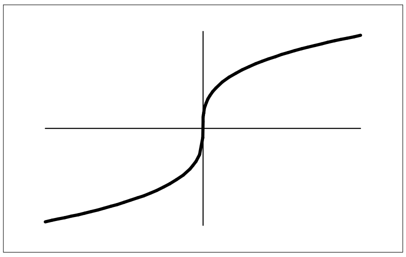
6. If $n = 2, 4, \ldots$ the shape of the graph $y = x^{\frac{1}{n}} = \sqrt[n]{x}$ is similar to the graph of $y = \sqrt{x}$.
7. If $n = 3, 5, \ldots$ the shape of the graph $y = x^{\frac{1}{n}} = \sqrt[n]{x}$ is similar to the graph of $y = \sqrt[3]{x}$.



$y = x^{\frac{1}{n}}, \; n = 2, 4, 6, \ldots$



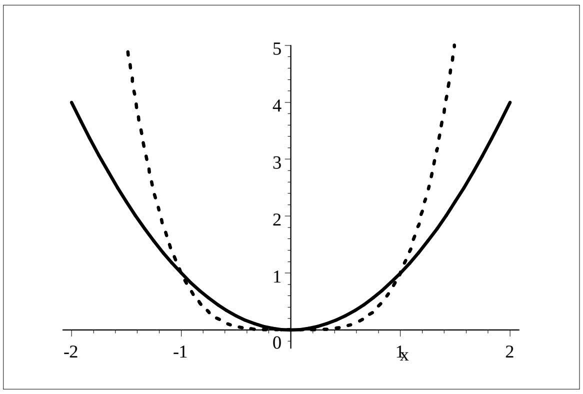
$y = x^{\frac{1}{n}}, \; n = 3, 4, 7, \ldots$

Example

Sketch $y = x^2$ and $y = x^4$ on the same set of axis.

The shapes are the same, but the graph with the higher power dominates. Dominates means that $x^4 > x^2$ for "large" values of x.

The dotted graph is $y = x^4$.



Definition (p 28)

The function $y = P(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$ is call a **polynomial.**

The degree of the polynomial is n (assuming that $a_n \neq 0$).

Examples of polynomials:

$y = ax^2 + bx + c$ (a quadratic polynomial if $a \neq 0$) and $y = ax^3 + bx^2 + cx + d$ (a cubic polynomial if $a \neq 0$).

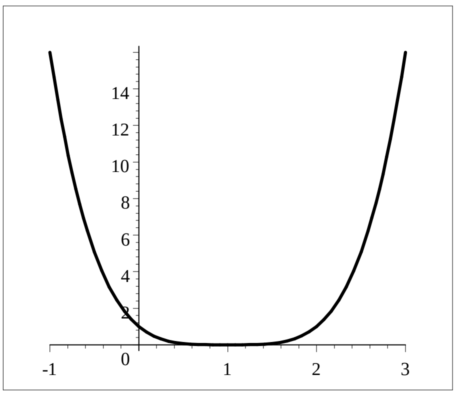
Remark

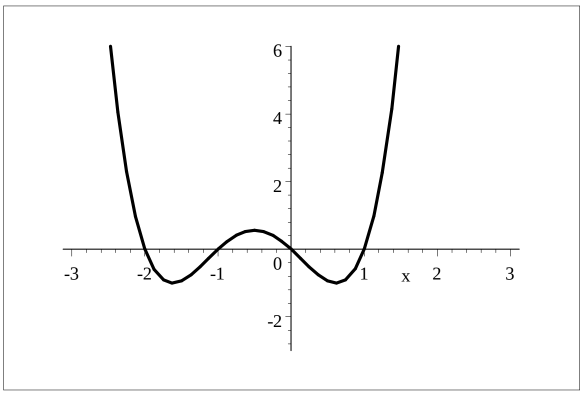
A polynomial of degree n has at most $n - 1$ turning points. Such a polynomial can have less than $n - 1$ turning points.

Example

Let $f(x) = x^4 - 4x^3 + 6x^2 - 4x + 1 = (x-1)^4$

and $g(x) = x^4 + 2x^3 - x^2 - 2x = x(x+1)(x-1)(x+2)$





$y = x^4 - 4x^3 + 6x^2 - 4x + 1$ $\qquad$ $y = x^4 + x^3 - 2x^2$

Both functions are polynomials of degree 4. The first one has one turning point at $x = 1$.

The second has three turning points.

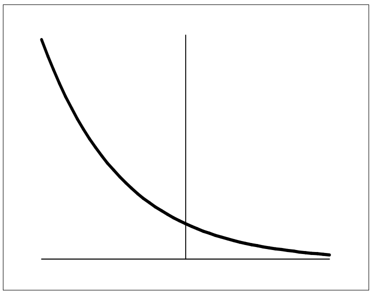
Later we will discuss the technique of finding the turning points.

Definition(p 31)
A **Rational function** is the ration of two polynomials.

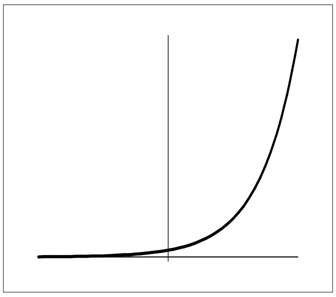
$y = R(x) = \dfrac{P(x)}{Q(x)}$ with $P(x)$ and $Q(x)$ polynomials.

Example
$y = \dfrac{x-1}{x^3+2x-1}$ is a rational function.
We will sketch these functions at a later stage.

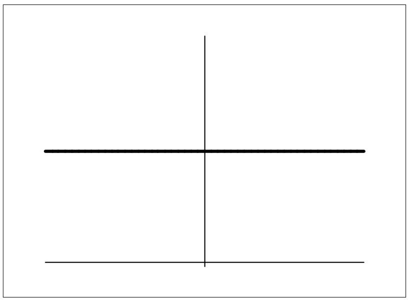
Definition (p 33)
The function $y = f(x) = a^x, \ a > 0, \ x \in \mathbb{R}$ is called an **exponential function**.



$y = a^x, \ 0 < a < 1$



$y = a^x, \ a > 1$



$y = a^x, \ a = 1$

Definition (p 34)
The function $y = f(x) = \log_a x$, with a a positive number is called a logarithmic function and is the inverse of an exponential function.